

Why Deep Learning? From Features to Representations

Scaler School of Technology | Deep Learning I | Week 1 — Session 1/1 | Post-Lecture Review & Practice



Key Takeaways

1 Hand-crafted features don't scale

Feature engineering (SIFT, HOG, BOW) works for tabular data but breaks down for images, audio, and text. Classical ML plateaus as data grows because the features are the ceiling.

2 Neural networks can learn anything (UAT)

The Universal Approximation Theorem guarantees a network with enough neurons can approximate any continuous function. But it doesn't tell you how many neurons, how to train, or how much data you need.

3 Representation learning is the paradigm shift

Deep networks learn hierarchical features automatically: edges → textures → parts → objects. These learned representations transfer across domains (ImageNet features work for medical imaging). Nobody programs this — it emerges from end-to-end training.

4 Know when to use what

DL dominates large unstructured data (images, text, audio). Classical ML wins on small tabular data, when interpretability matters, or when compute is limited. The 2×2 framework: data size vs. data type.

Test Your Understanding

Try answering these without looking at the slides. Answers are on page 3.

- Q1.** Explain why adding more training data helps deep learning but eventually stops helping classical ML with fixed features.
- Q2.** A colleague says “The UAT proves neural networks are always better than SVMs.” What's wrong with this statement?
- Q3.** You have 500 labeled chest X-rays and need to build a classifier. Would you use classical ML or deep learning? Justify your choice.
- Q4.** Why was AlexNet's victory in 2012 surprising, given that none of its individual ideas were new?

Work through these in a Colab notebook. No starter code — figure out the approach from what you learned in the lecture.

Exercise 1: Feature Engineering Showdown

Load CIFAR-10, extract HOG features from the images, and train a Random Forest. Then train a simple CNN (2 conv layers + FC) on the raw pixels. Compare accuracy. Which wins and by how much? Try with 1K, 5K, and 50K training samples — does the gap change?

Exercise 2: UAT in Action

Create a synthetic 1D function: $f(x) = \sin(3x) + 0.5 \cos(7x)$ for $x \in [-\pi, \pi]$. Train single-hidden-layer networks with 1, 5, 20, and 100 neurons. Plot all four approximations on the same graph. At what point does the fit become “good enough”?

Exercise 3: Visualize Learned Representations

Load a pre-trained VGG16 model. Extract features from 500 CIFAR-10 images at layers `features[5]`, `features[15]`, and `features[28]`. Use t-SNE to reduce each to 2D and plot with class colors. How does class separation improve in deeper layers?

Challenge Problem

The Scaling Experiment

Design an experiment that produces the “Andrew Ng plot” from the lecture. Use CIFAR-10 with training set sizes of 100, 500, 1K, 5K, 10K, and 50K. For each size, train: (a) a Random Forest on HOG features, and (b) a small CNN on raw pixels. Plot test accuracy vs. training set size for both. At what data size does the CNN overtake the Random Forest? Write a 3-sentence conclusion explaining your results.

- **Goodfellow, Bengio & Courville — Deep Learning, Ch. 1** — deeplearningbook.org
The definitive textbook intro. Covers exactly what we discussed today.
- **Krizhevsky et al. (2012) — ImageNet Classification with Deep CNNs** — the original AlexNet paper
Short, readable. The paper that started the deep learning revolution.
- **Zeiler & Fergus (2013) — Visualizing and Understanding CNNs** — arxiv.org/abs/1311.1901
The visualization paper we discussed — see feature maps at each layer.
- **LeCun, Bengio & Hinton (2015) — Deep Learning** — Nature review, nature.com/articles/nature14539
10-page field overview by three pioneers.

Online Resources & Tools

- **TensorFlow Playground** — interactive neural net visualization in your browser
- **3Blue1Brown Neural Networks** — YouTube series with excellent visual intuition
- **CNN Explainer** — interactive CNN visualization (poloclub.github.io/cnn-explainer)
- **PyTorch Tutorials** — pytorch.org/tutorials for hands-on practice
- **Distill.pub** — beautifully illustrated ML research articles
- **Google Colab** — free GPU access for all exercises

Fill the Gaps — Prerequisite Refreshers

Felt lost on a specific concept? These resources will help you catch up.

- **Feature engineering** — Review scikit-learn's `sklearn.feature_extraction` docs and your Classical ML notes
- **Overfitting/generalization** — Andrew Ng's ML Coursera, Week 3 (bias-variance tradeoff)
- **Linear algebra** — 3Blue1Brown "Essence of Linear Algebra" YouTube playlist
- **Python/NumPy** — NumPy quickstart tutorial at numpy.org
- **Gradient descent** — Preview: Andrew Ng's ML Coursera, Week 1 (covered in depth in our Week 2)

Answers — Test Your Understanding

- A1.** Fixed features are a ceiling. More data can't improve representations that are already maxed out. DL learns richer representations from more data — its features improve as data grows.
- A2.** The UAT says a solution *exists*, not that it's easy to find. It doesn't guarantee better performance, fewer parameters, or faster training than SVMs. In practice, you also need enough data and compute.
- A3.** 500 images is small — classical ML with hand-crafted features or **transfer learning** (fine-tuning a pre-trained CNN) would be best. Training a CNN from scratch on 500 images would overfit badly.
- A4.** The innovation was the **combination**: deep architecture + ReLU + GPU training + dropout + large data. No single component was new, but integrating them all enabled a breakthrough.

Hands-on notebook: www.priyanshsaxena.com/sst-deep-learning-nn

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